```
clear, clc

% 1. DATA INGESTION


% Creates a datastore object to manage image files in the specified
directory.
% Parameters: Folder path; 'FileExtensions' filters for specific image
formats.
imds = imageDatastore('C:\Users\Trent\OneDrive - University of
Wyoming\25-26\Image Processing\Test Images 1', 'FileExtensions', {'.jpg',
'.png', '.jpeg','.jfif'});

% 2. BATCH PROCESSING LOOP


% Iterates through the datastore as long as there are unread images
remaining.
while hasdata(imds)

    % Reads the current image and its metadata (info) from the datastore.
    [img, info] = read(imds);

    % Converts the image to grayscale
    img_gray = im2gray(img);


    % 3. PRE-PROCESSING & NOISE REDUCTION

    % Creates a disk-shaped structuring element with a 40-pixel radius for
morphological ops.
    se = strel('disk', 40);

    % Performs top-hat filtering to remove uneven background illumination
(shading correction).
    img_noshadow= imtophat(img_gray, se);

    % Applies anisotropic diffusion filtering to reduce noise while
preserving edge sharpess.
    img_denoised = imdiffusefilt(img_gray);

    % Performs Contrast Limited Adaptive Histogram Equalization (CLAHE) to
boost local contrast.
    % Parameter: 'ClipLimit' (0.001) prevents over-amplification of noise in
uniform areas.
    img_adjusted = adapthisteq(img_denoised,'ClipLimit',.001);

    % 4. FEATURE EXTRACTION (CRACK DETECTION)


    % Uses a Frangi vesselness filter to highlight tubular/line-like
structures (cracks).
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    % Parameters: imcomplement flips colors (cracks become light);
'Thickness' [1 8] sets the width range of cracks to detect;
'StructureSensitivity' (3) controls filter responsiveness.
    crackprob = fibermetric(imcomplement(img_adjusted), 'Thickness', [1 8],
'StructureSensitivity', 3);

    % Thresholding: Zeroes out any pixels with a probability lower than 0.3.
    crackprob(crackprob<.3) = 0;

    % Normalizes the probability map values to the range [0, 1].
    crackprob = rescale(crackprob);

    % 5. SEGMENTATION & MORPHOLOGY


    % Computes a locally adaptive threshold map using a Gaussian window.
    % Parameters: 0.6 sensitivity (higher = more foreground); 'Statistic'
sets the window mean type.
    T = adaptthresh(crackprob, 0.6, 'Statistic', 'gaussian');

    % Converts the grayscale probability map to a binary mask using the
adaptive threshold T.
    bw = imbinarize(crackprob,T);

    % Performs morphological closing (dilation then erosion) to bridge small
gaps in the cracks.
    bw_closed = imclose(bw, strel('disk',4));

    % Fills any isolated dark "holes" inside the detected binary crack
regions.
    bw_filled = imfill(bw_closed,"holes");

    % Performs morphological opening (erosion then dilation) to remove tiny
pixel artifacts.
    bw_final =  imopen(bw_filled,strel('disk',1));

    % Removes all connected components (objects) smaller than 200 pixels to
eliminate noise.
    final = bwareaopen(bw, 200);

    % 6. VISUALIZATION & ANALYSIS


    % Blends the detected crack mask onto the original color image for
visual verification.
    % Parameter: "Colormap" [0 0 1] displays the overlay in blue.
    overlay = labeloverlay(img, final,"Colormap",[0 0 1]); % overlays the
mask onto the original image

    % Calculates the total number of white (foreground) pixels to estimate
crack area.
    totalArea = sum(nnz(bw));

    % Displays the resulting overlay in a new figure window.
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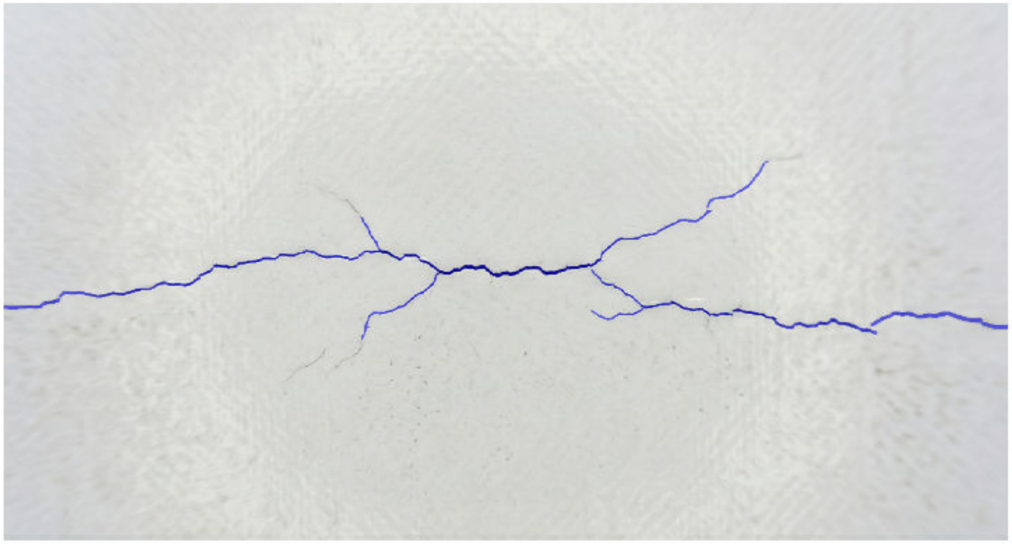
```
    figure
    montage({overlay})

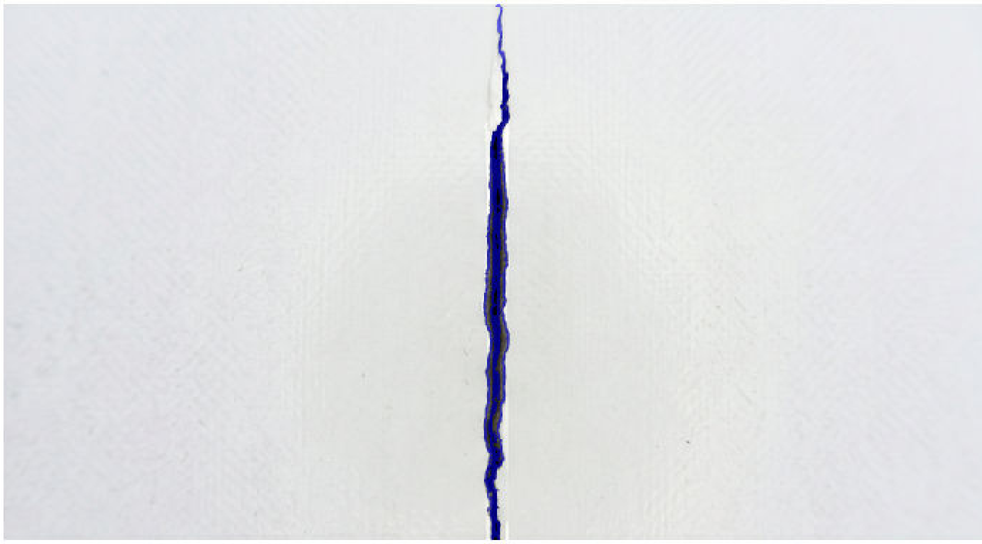
end


totalArea
% this is in pixels. Needs to be scaled to calculate area in in^2
```

```
totalArea =

       10331
```





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